

Transform Method for the Calculation of Vector-Coupled Sums: Application to the Spectral Form of the Vorticity Equation

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ABSTRACT

A transform method is developed for the fast calculation of vector-coupled sums appearing in the spectrally truncated vorticity equation. The method involves a kind of finite convolution theorem for expansions in surface harmonics. The method succeeds because it turns out to be much faster to transform the spectral representation to physical space, multiply the physical-space functions, and then inverse transform back to the spectral representation, than to evaluate the vector-coupled sums directly in spectral form. Direct evaluation of the sums requires order N^3 operations when the spectral representation is truncated at surface harmonics of degree N , while the transform method requires about $10N^3 + 40N^2 \log_2 N$ real operations per time step. Direct evaluation of the sums also requires storage of order N^3 , while the transform method requires storage of roughly $3N^3$ real words. An explanation is also given how to specialize the transform method to the "hemispheric" models of Baer and Platzman with a savings of nearly a factor of 8 in speed. For the hemispheric model with $N=19$, the transform method requires at least three times fewer calculations than are required by direct evaluation. The transform method is comparable in speed with direct evaluation of the vector-coupled sums when $N=10$ ($N=15$ for the hemispheric models), with the transform method improving its advantage over direct evaluation as N^2 . These new methods suggest that spectral representations with a large number of retained modes may be in a not too unfavorable competitive position with finite-difference jury methods (which require at least order $N^3 \log_2 N$ operations per time step) for the solution of the equations of incompressible spherical hydrodynamics.

1. Introduction

Truncated spectral representations have a number of important advantages over finite-difference methods for accurate numerical simulation of two-dimensional hydrodynamic flows on the surface of a sphere. Platzman (1960, 1962), Baer and Platzman (1961) and Ellsaesser (1966) emphasize three advantages of the spectral form of simulation: 1) a mapping of the sphere is unnecessary, so that the known difficulties of finite-difference methods near the poles are eliminated; 2) the interaction coefficients between retained modes are treated exactly, so that the truncated dynamical system conserves energy and enstrophy by nonlinear interaction and no instabilities attributable to aliasing errors can exist; and 3) the planetary-wave components of the motion can be treated exactly (in both space and time) so that there are no phase-speed errors. The principal disadvantage of the spectral representation is the large amount of arithmetic necessary to calculate spectral equations with even a moderate number of retained modes. This is apparently the reason why numerical simulations of atmospheric motions are mostly done with finite-difference methods. If the spectral repre-

sentation is truncated at surface harmonics of degree N (see Section 2), then there are $(N+1)^2$ retained *real* modes. With previously suggested methods to compute the spectral form of the vorticity equation, the number of arithmetic calculations per time step is of order N^5 .

In this paper, we present a method of computing the spectral form of the vorticity equation that reduces the number of arithmetic operations per time step to order N^3 . The method involves use of a kind of convolution theorem for the vector-coupled sums appearing in the truncated spectral equations. This new method is an extension to truncated surface-harmonic representations of a method developed for truncated Fourier-series representations (Orszag, 1969). The latter method involves efficient evaluation of alias-free convolution sums appearing in the spectrally transformed three-dimensional Navier-Stokes equations by means of the fast Fourier transform (Cooley and Tukey, 1965) and a finite form of the convolution theorem. An exact analog of the fast Fourier transform is not available for the present application, but, for N not too small, it still turns out to be much more efficient to transform, multiply, and inverse transform than it is to "convolve," i.e., compute the vector-coupled sums directly.

Finite-difference jury methods require at least order $N^3 \log_2 N$ operations per time step on an $N \times N$ grid. However, most finite-difference schemes are not this

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efficient, since the relaxation methods for solution of the Poisson equation for the stream function that are usually used require $O(N^3)$ operations. Consequently, the method of spectral representation can no longer be dismissed outright as a wasteful computation at moderately large N . The fundamental advantages of the spectral form suggest that the new method developed in this paper be considered seriously as a possible replacement for finite-difference simulations. Further study will be necessary, however, to determine the conditions under which such a replacement would be warranted.

2. The truncated spectral form of the vorticity equation

The vorticity $\zeta(\theta, \phi)$, where θ is the co-latitude and ϕ the longitude in a spherical coordinate system, is approximated by a truncated expansion in surface harmonics of the form

$$\zeta(\theta, \phi) = \sum_{n=0}^N \sum_{|m| \leq n} a_{nm} Y_n^m(\theta, \phi). \quad (1)$$

Here the surface harmonic Y_n^m of degree n and order m is defined by

$$Y_n^m(\theta, \phi) = \left[\frac{(2n+1)(n-m)!}{4\pi(n+m)!} \right]^{\frac{1}{2}} P_n^m(\cos\theta) e^{im\phi}, \quad (2)$$

where the associated Legendre function $P_n^m(x)$ is given by

$$P_n^m(x) = \frac{(-1)^m}{2^n n!} (1-x^2)^{m/2} \frac{d^{n+m}}{dx^{n+m}} (x^2-1)^n, \quad |m| \leq n. \quad (3)$$

The definitions (2) and (3) imply that

$$Y_n^{-m}(\theta, \phi) = (-1)^m [Y_n^m(\theta, \phi)]^*, \quad (4)$$

$$\int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta [Y_n^m(\theta, \phi)]^* Y_p^r(\theta, \phi) = \delta_{n,p} \delta_{m,r}, \quad (5)$$

$$\int_{-1}^1 P_n^m(x) P_p^m(x) dx = \delta_{n,p} \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}, \quad (6)$$

where the asterisk denotes complex conjugate and $\delta_{n,p} = 1$ if $n=p$, and is zero otherwise. It follows from (5) that the coefficients a_{nm} in (1) are given by

$$a_{nm} = \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta \zeta(\theta, \phi) [Y_n^m(\theta, \phi)]^*. \quad (7)$$

The surface harmonics Y_n^m satisfy

$$\nabla^2 Y_n^m = -n(n+1) Y_n^m \quad (8)$$

on the surface of the unit sphere. Therefore, the stream-function $\psi(\theta, \phi)$ associated with the vorticity $\zeta = \nabla^2 \psi$ is

$$\psi(\theta, \phi) = \sum_{n=1}^N \sum_{|m| \leq n} b_{nm} Y_n^m(\theta, \phi), \quad (9)$$

where $b_{nm} = -a_{nm}/(n^2+n)$. Reality of ζ, ψ on the surface of the sphere requires that $a_{nm}^* = (-1)^m a_{n,-m}$.

The vorticity equation for nondivergent planetary flow in a reference frame rotating with angular velocity Ω is

$$\frac{\partial \zeta}{\partial t} = -2\Omega \frac{\partial \psi}{\partial \phi} + J(\theta, \phi), \quad (10)$$

where the Jacobian $J(\theta, \phi)$ is given by

$$J(\theta, \phi) = \frac{\partial \zeta}{\partial \phi} \frac{\partial \psi}{\partial \cos\theta} - \frac{\partial \zeta}{\partial \cos\theta} \frac{\partial \psi}{\partial \phi}. \quad (11)$$

The truncated spectral form of (10) is obtained by substituting the expressions (1) and (9) into (10) and (11), multiplying the result by $[Y_n^m(\theta, \phi)]^*$ ($0 \leq |m| \leq n \leq N$), and integrating over the surface of the unit sphere to find an equation for a_{nm} ($0 \leq |m| \leq n \leq N$). The result is

$$\frac{\partial}{\partial t} a_{nm} = -2i\Omega m c_n a_{nm} + J_{nm}, \quad (12)$$

where $c_n = 1/(n^2+n)$ and

$$J_{nm} = \sum_{p=0}^N \sum_{q=0}^N \sum_{|r| \leq N} I(p, r, q, m-r | n, m) a_{pr} b_{q, m-r} \quad (0 \leq |m| \leq n \leq N), \quad (13)$$

with the interaction coefficient I given by

$$I(p, r, q, m-r | n, m) = \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta [Y_n^m(\theta, \phi)]^* \times \left[\frac{\partial}{\partial \phi} Y_p^r(\theta, \phi) \frac{\partial}{\partial \cos\theta} Y_q^{m-r}(\theta, \phi) - \frac{\partial}{\partial \cos\theta} Y_p^r(\theta, \phi) \frac{\partial}{\partial \phi} Y_q^{m-r}(\theta, \phi) \right]. \quad (14)$$

The relation $b_{nm} = -c_n a_{nm}$ allows some simplification of the sum (13), but we shall consider the general form of the Jacobian in which a_{nm} and b_{nm} have no *a priori* relation. The interaction coefficient I is zero unless

$r \neq 0$ or $m \neq 0$; $npq \neq 0$; $p \neq q$ or $r \neq m-r$;

and $|r| \neq p$ or $|m-r| \neq q$; (15a)

$|p-q| < n < p+q$; (15b)

$p+q+n$ is an odd integer; (15c)

$|r| \leq p$; $|m-r| \leq q$. (15d)

The selection rules (15) significantly restrict the range of summation in (13). However, it is apparent from (13) that if the truncation is $0 \leq |m| \leq n \leq N$, the total number of operations involved in calculating (13) for all retained n, m scales as N^5 . A Monte Carlo evaluation of the probability of the selection rules (15) being satisfied shows that the number of nonzero terms involved in (13) is $\sim 0.226N^5$ for N large. Each of the supplementary conditions $a_{nm}^* = (-1)^m a_{n,-m}$ and $b_{nm} = -c_n a_{nm}$ reduces the number of terms actually involved in (13) by about a factor of 2. However, this total reduction by a factor of 4 is partially compensated by the fact that a_{nm} is complex and I is pure imaginary so that the arithmetic is increased by a factor of 2.

Platzman (1960) observed that a closed subsystem of (12) is obtained by dropping all a_{nm} with $n+m$ even. If all a_{nm}, b_{nm} with $n+m$ even are zero initially, then the selection rule (15c) written as $(p+r) + (q+m-r) + (n+m) = \text{odd integer}$ implies that J_{nm} with $n+m$ even is zero initially. It is then possible to show by induction that all time derivatives of a_{nm} with $n+m$ even are zero, so that no nonzero a_{nm} with $n+m$ even develops in time evolution according to (12). These closed subsystems involve only a_{nm} with $n+m$ odd, so that the physical-space vorticity and streamfunction are both antisymmetric with respect to the equator. In their numerical studies, Baer and Platzman (1961) and Ellsaesser (1966) considered only these "hemispheric" models with a_{nm} zero for $n+m$ even. Baer and Platzman counted the minimum number of terms needed to compute (13) directly accounting for the selection rules (15), the reality condition $a_{nm}^* = (-1)^m a_{n,-m}$, and the relation $b_{nm} = -c_n a_{nm}$. They found 38,128 terms in the hemispheric model with the slightly modified cutoff $2 \leq n \leq 19$, $|m| \leq n$, and 1,026,069 terms with the cutoff $2 \leq n \leq 37$, $|m| \leq n$. The scaling on N^5 is apparent. Since each term in (13) involves at least one multiplication of a complex number by a pure imaginary number followed by a complex addition, at least 75,000 real operations are required to calculate (13) when $N=19$, and at least 2,000,000 real operations are required when $N=37$. An operation is defined as a multiplication followed by an addition. These estimates of operations for direct calculation of (13) in the hemispheric models are conservative. In fact, the number of arithmetic operations involved is somewhat larger because of the need to compute complex products such as $a_{pr} a_{q,m-r}$. A computer count indicates that the minimum number of real operations to compute all the needed products of a_{nm} 's (corresponding to nonzero interaction coefficients) is roughly 18,000 for $N=19$ and 250,000 for $N=37$. Finally, the Monte Carlo calculation cited above shows that the number of real operations involved in the hemispheric models is roughly $0.028N^5$ per time step, when N is large. The hemispheric models are discussed further in Section 4. In the next section, we present the transform method for the general model.

3. Transform method for calculating J

It is clear that the only non-trivial calculational task associated with the truncated spectral equation (12) is the evaluation of the Jacobian (13). The method presented in this section reduces the number of arithmetic operations from $O(N^5)$ to $O(N^3)$ at the same time that it reduces the necessary storage from $O(N^5)$ [for all the interaction coefficients I] to $O(N^3)$.

Consider the truncated spectral representation of ζ, ψ given by (1) and (9), respectively. Each of these representations involves surface harmonics of degree at most N (or, in the language of quantum mechanics, total angular momenta of at most N). For these truncated ζ, ψ , the Jacobian defined by (11) is *exactly*

$$J_N(\theta, \phi) = \sum_{p=0}^N \sum_{|r| \leq p} \sum_{q=0}^N \sum_{|s| \leq q} a_{pr} b_{qs} Z_{pq}^{rs}(\theta, \phi), \quad (16)$$

where

$$Z_{pq}^{rs}(\theta, \phi) = \frac{\partial}{\partial \phi} Y_p^r(\theta, \phi) \frac{\partial}{\partial \cos \theta} Y_q^s(\theta, \phi) - \frac{\partial}{\partial \cos \theta} Y_p^r(\theta, \phi) \frac{\partial}{\partial \phi} Y_q^s(\theta, \phi). \quad (17)$$

The crucial fact that permits the technique described in this paper is that Z_{pq}^{rs} is expandable in a *truncated* surface-harmonic expansion

$$Z_{pq}^{rs}(\theta, \phi) = \sum_{n=|r+s|}^{p+q-1} I(p, r, q, s | n, r+s) Y_n^{r+s}(\theta, \phi), \quad (18)$$

where $I(p, r, q, s | n, r+s)$ is given generally by (14) [not just when $n \leq N$]. We postpone the proof of (18) to the next paragraph. Assuming (18), it is apparent that

$$J_{nm} = \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta [Y_n^m(\theta, \phi)]^* J_N(\theta, \phi), \quad (19)$$

for $0 \leq |m| \leq n \leq N$. Examining the derivation of (12), it is apparent that (19) is really just the definition of J_{nm} , but the formal analysis leading to (19) will be useful below.

The proof of Eq. (18) is straightforward. We can assume that $r+s \geq 0$, since the result for $r+s < 0$ follows by complex conjugation. Each $P_n^m(x)$ is of the form

$$(1-x^2)^{m/2} p_{n-m}(x), \quad |m| \leq n,$$

where $p_r(x)$ is a generic label for a polynomial in x of degree r . Consequently,

$$\left. \begin{aligned} \frac{\partial}{\partial \cos \theta} Y_n^m(\theta, \phi) &= -\frac{m \cos \theta}{1 - \cos^2 \theta} Y_n^m(\theta, \phi) \\ &\quad + e^{im\phi} (1 - \cos^2 \theta)^{m/2} p_{n-m-1}(\cos \theta) \\ \frac{\partial}{\partial \phi} Y_n^m(\theta, \phi) &= im Y_n^m(\theta, \phi) \end{aligned} \right\}, \quad (20)$$

where $p_{n-m-1}(x)$ is a symbol for a polynomial in x of degree $n-m-1$. Therefore,

$$Z_{pq}^{rs}(\theta, \phi) = e^{i(r+s)\phi} \times (1 - \cos^2 \theta)^{(r+s)/2} p_{p+q-r-s-1}(\cos \theta), \quad (21)$$

since the terms involving $[m \cos \theta / (1 - \cos^2 \theta)]$ in (20) cancel when substituted in (17). Now every polynomial of degree at most $p+q-r-s-1$, in particular $p_{p+q-r-s-1}(x)$, can be expanded in terms of the polynomials $q_n^{r+s}(x) = (1-x^2)^{-(r+s)/2} P_n^{r+s}(x)$ for $r+s \leq n \leq p+q-1$, since $q_n^{r+s}(x)$ is of degree exactly $n-r-s$ when $n \geq r+s \geq 0$. This proves that a *truncated* expansion of the form (18) for Z_{pq}^{rs} exists, but does not yet establish that the coefficients are given by (14). However, once it is known that the expansion (18) truncates at $n=p+q-1$, the coefficients must be given by (14) as shown by a simple generalization of (7).

The important result of (18) is that $J_N(\theta, \phi)$ given by (16) is expansible in the truncated series

$$J_N(\theta, \phi) = \sum_{n=0}^{2N-1} \sum_{|m| \leq n} d_{nm} Y_n^m(\theta, \phi), \quad (22)$$

where $d_{nm} = J_{nm}$ for $0 \leq |m| \leq n \leq N$. However, the coefficients d_{nm} may be computed *exactly* without computing either the integral (19) or the sums (13). The method is to choose at least $2N$ values of θ , say $\theta_1, \theta_2, \dots, \theta_{2N}$, distributed between 0 and π but not including either 0 or π , and also $4N$ values of ϕ , say $\phi_1, \phi_2, \dots, \phi_{4N}$, distributed between 0 and 2π . The values of $J_N(\theta, \phi)$ at these $8N^2$ points may be computed directly from the coefficients a_{nm}, b_{nm} by computing the finite transforms

$$F_1(\theta_j, \phi_k) = \sum_{n=0}^N \sum_{|m| \leq n} a_{nm} \frac{\partial}{\partial \cos \theta} Y_n^m(\theta_j, \phi_k), \quad (23)$$

$$F_2(\theta_j, \phi_k) = \sum_{n=0}^N \sum_{|m| \leq n} im b_{nm} Y_n^m(\theta_j, \phi_k), \quad (24)$$

$$F_3(\theta_j, \phi_k) = \sum_{n=0}^N \sum_{|m| \leq n} b_{nm} \frac{\partial}{\partial \cos \theta} Y_n^m(\theta_j, \phi_k), \quad (25)$$

$$F_4(\theta_j, \phi_k) = \sum_{n=0}^N \sum_{|m| \leq n} ima_{nm} Y_n^m(\theta_j, \phi_k), \quad (26)$$

for $j=1, \dots, 2N$ and $k=1, \dots, 4N$. The method to compute the transforms (23)–(26) is discussed later. In terms of the transforms, the Jacobian becomes

$$J_N(\theta_j, \phi_k) = F_1(\theta_j, \phi_k) F_2(\theta_j, \phi_k) - F_3(\theta_j, \phi_k) F_4(\theta_j, \phi_k), \quad (27)$$

for $j=1, \dots, 2N$ and $k=1, \dots, 4N$.

Eq. (22) implies that

$$J_N(\theta_j, \phi_k) = \sum_{n=0}^{2N-1} \sum_{|m| \leq n} d_{nm} X_{nj}^m e^{im\phi_k}, \quad (28)$$

where

$$Y_n^m(\theta_j, \phi_k) = X_{nj}^m e^{im\phi_k}.$$

Now choose $\phi_k = 2\pi k / (4N)$ so that

$$\sum_{n=|m|}^{2N-1} d_{nm} X_{nj}^m = g_m(\theta_j), \quad (m = -N, -N+1, \dots, N), \quad (29)$$

where $g_m(\theta_j)$ is defined as

$$g_m(\theta_j) = \frac{1}{4N} \sum_{k=1}^{4N} J_N(\theta_j, \phi_k) e^{-2\pi i m k / 4N}. \quad (30)$$

Eq. (29) follows from (28) and (30) because

$$\frac{1}{4N} \sum_{k=1}^{4N} e^{2\pi i (m-n) k / 4N} = \delta_{m,n}, \quad (31)$$

which is the fundamental property of the finite Fourier transform. Notice that (29) involves $2N+1$ separate equations, one for each value of m . Define $\bar{X}_{jn}^m (j=1, \dots, 2N-|m|, n=|m|, \dots, 2N-1, m=-N, -N+1, \dots, N)$ so that

$$\sum_{j=1}^{2N-|m|} X_{nj}^m \bar{X}_{jp}^m = \delta_{n,p}, \quad (n=|m|, \dots, 2N-1), \quad (32)$$

for all $|m| \leq N$. The existence of the “inverse” matrices $\bar{X}$ is assured by the integral property (6), because (32) is a discrete analog of (6), as long as the points $\theta_j, j=1, \dots, 2N$, are chosen wisely. That is, the special sets of points θ_j for which $\bar{X}$ does not exist is anticipated to be of zero relative measure because of (6), so that $\bar{X}$ exists for almost all sets of θ_j . Once $\bar{X}$ is found, it follows from (29) and (32) that

$$d_{nm} = \sum_{j=1}^{2N-|m|} g_m(\theta_j) \bar{X}_{jn}^m, \quad (33)$$

so that

$$J_{nm} = \sum_{j=1}^{2N-|m|} g_m(\theta_j) \bar{X}_{jn}^m, \quad (0 \leq |m| \leq n \leq N). \quad (34)$$

The peculiar choice of $\bar{X}_{jn}^m$ to be the partial inverse matrix of X_{nj}^m restricted to the first $2N-|m|$ values of j and the last $2N-|m|$ values of n causes no difficulty. This definition of $\bar{X}$ is just an artifice to determine the d_{nm} . The representation (28) ensures that the same values of d_{nm} are obtained wherever the points θ_j are located (as long as $\bar{X}$ exists) and whichever $2N-|m|$ of the θ_j are used to invert X_{nj}^m .

The method proposed here is to calculate the vector-coupled sum (13) by first calculating the transforms (23)–(26), then constructing J_N by (27), then g_m by (30), and finally finding the sum (13) by (34). It remains to determine the number of arithmetic operations involved in each of these steps. Before proceeding to do this, it is useful to emphasize again the basic idea behind the present method. The point is that the vector-coupled sums (13) involve much computation because of their “non-local” nature. The transforms (23)–(26)

convert the non-local sum (13) into the "local" multiplication in physical space expressed in (27). The speed of the present method owes to the fact that transforming to θ, ϕ space [(23)–(26)], local multiplication [(27)], and inverse transformation [(30), (34)] is much faster than direct computation of (13).

An *a priori* estimate of the number of arithmetic computations necessary to compute (23)–(26) is $8N^4$ due to the $2N$ values of j , the $4N$ values of k , and the N^2 pairs m, n with $0 \leq |m| \leq n \leq N$. This estimate may be reduced considerably. We will explain the method only for F_1 given by (23), but closely similar methods work also for F_2, F_3, F_4 with the same final number of computations. Define W_{nj}^m by

$$\frac{\partial}{\partial \cos \theta} Y_n^m(\theta_j, \phi_k) = W_{nj}^m e^{im\phi_k},$$

and set

$$h_m(\theta_j) = \sum_{n=|m|}^N a_{nm} W_{nj}^m, \quad (35)$$

for $j=1, \dots, 2N$, $m=-N, -N+1, \dots, N$. Notice that (4) implies that $W_{nj}^{-m} = (-1)^m W_{nj}^m$ is a real number, so that $a_{nm}^* = (-1)^m a_{n,-m}$ gives

$$[h_{-m}(\theta_j)]^* = h_m(\theta_j). \quad (36)$$

Therefore, $h_m(\theta_j)$ need only be computed for $m=0, \dots, N$. Since each W_{nj}^m is real while each a_{nm} is complex, it follows that computation of (35) requires about $2N^3$ real multiplications and additions. Using (35), Eq. (23) may be computed as

$$F_1(\theta_j, \phi_k) = \sum_{m=-N}^N h_m(\theta_j) e^{2\pi i m k / (4N)}, \quad (37)$$

with the choice $\phi_k = 2\pi k / (4N)$ made earlier. The transform in (37) may be computed by the fast Fourier transform with about $4N \log_2 N$ real operations for each fixed value of j , so that (37) involves a total of about $8N^2 \log_2 N$ real operations. The fast Fourier transform may be applied to (37) either by extending the range of m to $-2N \leq m \leq 2N-1$ by appending zeroes to $h_m(\theta_j)$ (for each fixed j), or by splitting (37) up into two transforms with $-N \leq m \leq N-1$, one transform on even k and one on odd k . We have also included a factor $\frac{1}{2}$ in our estimate of calculations involved in (37), because the "half-complex" condition (36) is readily accounted for by the fast Fourier transform algorithm with a resulting speedup of nearly a factor of 2. The final result of (35) and (37) is that (23) involves about $2N^3 + 8N^2 \log_2 N$ real arithmetic operations. Thus, evaluations of (23)–(26) involve a total of about $8N^3 + 32N^2 \log_2 N$ real operations.

Evaluation of (27) involves about $16N^2$ real operations, since F_1, F_2, F_3, F_4 are pure real numbers. Also, evaluation of $g_m(\theta_j)$ for $|m| \leq N$, $j=1, \dots, 2N$, by the

fast Fourier transform involves about $8N^2 \log_2 N$ real operations. The resulting $g_m(\theta_j)$ is also half-complex, in the sense that $g_m(\theta_j) = [g_{-m}(\theta_j)]^*$. Finally, evaluation of J_{nm} by (34) for $0 \leq |m| \leq n \leq N$ requires about $(5/3)N^3$ real operations, since

$$\int_0^N (N-m)(2N-m) dm = \frac{5}{6} N^3,$$

while (34) need only be computed for $0 \leq m \leq n \leq N$ and $\bar{X}_{jn}^m$ is necessarily real. Adding up all these estimates concerning (23)–(26), (27), (30) and (34), the complete computation of J_{nm} for $0 \leq |m| \leq n \leq N$ using the method described above involves about $10N^3 + 40N^2 \times \log_2 N$ real operations. Also, the principal quantities that must be stored (in core, disk or drum memory) are the kernels W_{nj}^m of the transforms (23)–(26) and $\bar{X}_{jn}^m$ for $j=1, \dots, 2N-|m|$, $n=|m|, \dots, N$, $m=0, \dots, N$. Since the kernel of (23) is identical to that of (25) and the kernel of (24) is identical to that of (26), it follows that the total storage for the two W matrices and one $\bar{X}$ matrix is about $3N^3$.

4. Generalizations

The first generalization (or, rather, specialization) of the method of Section 3 that we shall discuss is to the hemispheric models mentioned in Section 2. In the hemispheric models, the only nonzero a_{nm} (and b_{nm}) have $n+m$ odd. In this case, the method of the last section may be speeded up by about a factor of 8. First of all, it is sufficient to evaluate the transforms (23)–(26) and the Jacobian (27) at only N values θ_j and $2N$ values ϕ_k . The N values θ_j may all be chosen in a single hemisphere, if desired. Also, the sums (35) need only be taken over even n when m is odd and odd n when m is even. Estimating the required amount of arithmetic as before gives the estimate $(5/4)N^3 + 10N^2 \log_2 N$ real operations per time step for the hemispheric models with cutoff at surface harmonics of degree N . The required storage is about N^3 .

The estimate just given for the number of operations involved in computing the sums (13) for the hemispheric model by the transform method may be compared to the number of operations involved in the direct evaluation of (13) as determined by Baer and Platzman. With $N=19$, there are 190 independent *real* modes in the hemispheric model. Direct evaluation of the sums (13) requires at least 75,000 real operations (cf. Section 2), while the transform method requires about 25,000 operations. With $N=37$, there are 703 real modes. Direct methods require at least 2,000,000 real operations to compute (13), while the transform method requires only about 140,000 real operations. (Note that we have made these estimates for the transform method using the known operation counts for the fast Fourier transform which are valid only when N is a power of 2.)

Ellsaesser (1966) considers the modified spectral truncation $|m| \leq N$, $0 \leq n - |m| \leq N$ instead of $0 \leq |m| \leq n \leq N$. The transform method can again be applied with the modification that $4N$ values of θ_j must be chosen together with $4N$ values of ϕ_k . In this case, the upper limit of summation in (29) is $4N-1$, while the upper limit of summation in (35) is $N+|m|$. Hemispheric models again give a closed subsystem with a_{nm} , $n+m$ even, all zero. An operation count like those above gives $(9/2)N^3 + 20N^2 \log_2 N$ real operations for Ellsaesser's hemispheric model (for large N equal to a power of 2). For each value of N , Ellsaesser's model includes about twice as many modes as the models with cutoff $0 \leq |m| \leq n \leq N$, so there is some relative loss of speed in the application of the transform methods to Ellsaesser's model, due principally to the fact that modes with $N+|m| \leq n \leq 2N$, $|m| \leq N$ are discarded. Even so, Ellsaesser counts that 145,559 terms must be evaluated to compute (13) for his hemispheric model with $N=18$. This gives at least 290,000 real operations per time step for direct evaluation of the Jacobian, while the transform method requires only about 55,000 real operations per time step.

Another simple generalization of the method involves computation of sums like

$$K_{nm} = \sum_{p=0}^N \sum_{q=0}^N \sum_{|r| \leq N} K(p, r, q, m-r | n, m) a_{pr} b_{q, m-r}, \quad (38)$$

where

$$K(p, r, q, m-r | n, m) = \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta \times [Y_n^m(\theta, \phi)]^* Y_p^r(\theta, \phi) Y_q^{m-r}(\theta, \phi). \quad (39)$$

A sum of the form (38) would result if the dynamical equation (10) included a term like $\zeta(\theta, \phi)\psi(\theta, \phi)$. The coupling coefficients K are simple multiples of vector-addition coefficients (Clebsch-Gordan coefficients, Wigner 3- j symbols)—hence, the reason for calling (13) and (38) vector-coupled sums. The transform method goes through almost exactly as before, except that $Z_{pq}^{rs}(\theta, \phi) = Y_p^r(\theta, \phi) Y_q^s(\theta, \phi)$ and (18) is replaced by the well-known law for addition of angular momenta. The transforms (23)–(26) are also changed in the obvious way. Again, the crucial property that justifies the whole process is the result analogous to (18), that $Z_{pq}^{rs}(\theta, \phi)$ can be expanded in a truncated series of $Y_n^{r+s}(\theta, \phi)$ for $r+s \leq n \leq p+q$. For (38), this is the well-known property that two angular momenta of magnitude at most N couple to give a total angular momentum of at most $2N$.

Finally, it should be noted that the transform method cannot be used to evaluate the truncated spectral form

of a term like

$$L(\theta, \phi) = \frac{\partial \zeta}{\partial \phi} \frac{\partial \psi}{\partial \cos \theta},$$

i.e., one of the terms of $J(\theta, \phi)$ taken separately. The reason is that

$$Z_{pq}^{rs}(\theta, \phi) = \frac{\partial}{\partial \phi} Y_p^r(\theta, \phi) \frac{\partial}{\partial \cos \theta} Y_q^s(\theta, \phi)$$

cannot always be expanded in a *truncated* expansion analogous to (18), because the first term of the right-hand side of (20) is no longer cancelled. In this case, it does not seem possible to eliminate exactly the aliasing interactions without direct evaluation of the sums analogous to (13). A representation of the form (28) for $L_N(\theta, \phi)$ is no longer ensured and “inversion” of $L(\theta, \phi)$ on any finite grid introduces aliases.

In conclusion, it seems that the transform method may make possible large-scale numerical simulations using the spectral form of the hydrodynamic equations. It seems that the break-even point with direct calculation of the vector-coupled sums is near the cutoff $N=15$ for hemispheric models and $N=10$ for general spectral truncation. The advantage of the transform method over the direct method increases roughly as N^2 . The transform method also enjoys the advantage that it requires much less total storage than do direct methods when $N \gtrsim 10$, so that programming should be easier. The transform method also saves the very bothersome calculation of the interaction coefficients $I(p, r, q, m-r | n, m)$ by replacing it by the construction of the “inverse” matrix $\bar{X}$ defined by (32). The calculation of $\bar{X}$ is straightforward using known methods (such as Gaussian elimination) for inverting matrices.

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